
Jânio Anselmo, M.Sc.

PhD Candidate in Electrical Engineering (Biomedical Engineering), UFSC
Electrical Engineer – CREA-SC 074576-3

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| Summary

PhD candidate in Electrical Engineering at the Federal University of Santa Catarina (UFSC), Brazil, with a M.Sc. in the same program (Biomedical Engineering concentration). Doctoral research covers mathematical modeling and experimental validation of pulsed electric fields (PEF) in continuous flow, combining numerical modeling, laboratory instrumentation and parameter estimation against measured data.

Co-founder and R&D Director of ENSA Tecnologia since 2009, leading a team of four and designing microcontroller-based electronic systems across biomedical signal acquisition, environmental sensing, industrial automation and pattern classification, with a Brazilian invention patent filed in 2025.

Published peer-reviewed work in numerical modeling of electroporation and biomedical systems. Served as a peer reviewer for the American Medical Informatics Association (AMIA), sat on three undergraduate thesis committees and taught the Transducers Laboratory course at UFSC in two assistantships — work that consists of explaining technical concepts precisely and judging technical output against explicit criteria.

| Education

Ph.D. Candidate, Electrical Engineering — Biomedical Engineering

2023–2027

Federal University of Santa Catarina (UFSC) · Florianópolis, Brazil

Graduate Program in Electrical Engineering (PPGEEL). Thesis: *Pulsed electric fields in continuous flow: effective electric dose modeling and experimental validation*. Advisor: Prof. Daniela Ota Hisayasu Suzuki, Ph.D.; co-advisor: Prof. Guilherme Brasil Pintarelli, Ph.D.

M.Sc., Electrical Engineering — Biomedical Engineering

2012–2014

Federal University of Santa Catarina (UFSC) · Florianópolis, Brazil

Dissertation: *Single-Cell Electroporation Study* — numerical study of the electrical and mechanical effects of capillary-electrode electroporation on the cell membrane.

B.Eng., Electrical Engineering — Telematics

2006–2011

University of Southern Santa Catarina (UNISUL) · Palhoça, Brazil

Electives in power electronics and computer networks.

Also: Postgraduate Specialization in Business Management, FMP (2015–2018) · B.Sc. Technology in Telecommunications Systems, IFSC (2008–2015) · Technical Diploma in Telecommunications, IFSC (1997–2001).

Research Experience

Doctoral Researcher — Pulsed Electric Fields in Continuous Flow

2023–2027

Federal University of Santa Catarina (UFSC) · Florianópolis, Brazil

- Developed an effective electric dose model for yeast (*Saccharomyces cerevisiae*) inactivation under continuous-flow PEF treatment, keeping pulse repetition rate and residence time as separate factors.
- Built and instrumented the experimental bench; acquired oscilloscope voltage and current waveforms, temperature records and conductivity measurements as the model's validation set.
- Estimated model parameters and assessed them out of sample, including leave-one-out validation and bootstrap confidence intervals for the fitted coefficients.
- Wrote the resulting manuscript in LaTeX under the IEEE Access template, in both English and Portuguese.

Master's Researcher — Single-Cell Electroporation

2012–2014

Federal University of Santa Catarina (UFSC) · Florianópolis, Brazil

Numerical study of electroporation of isolated biological cells through a stretched capillary electrode, covering the electrical and mechanical effects on the cell membrane. Advisors: Prof. Daniela O. H. Suzuki, Ph.D. and Prof. Jefferson L. B. Marques, Ph.D. In parallel, developed low-cost hardware for heartbeat and cardiac audio monitoring in the *Smart Medical Devices* course with Prof. Mohamad Sawan, P.Eng., Ph.D. (2013).

Publications

- **Anselmo, J.**, Suzuki, D. O. H., Pintarelli, G. B., Antônio Júnior, A. de C., Silva, J. R. da. *Mathematical Modeling and Experimental Assessment of Pulsed Electric Fields in Continuous Flow for Saccharomyces cerevisiae*. Submitted, under review, 2026.
- Suzuki, D. O. H., **Anselmo, J.**, de Oliveira, K. D., Freytag, J. O., Rangel, M. M. M., Marques, J. L. B., Ramos, A. *Numerical Model of Dog Mast Cell Tumor Treated by Electrochemotherapy*. Artificial Organs, 2014.
- Fronza, C. F., **Anselmo, J.**, Marques, J. L. B., Pintarelli, G. B., Castro, A. de. *A Blower-Like Approach to Predict the Effectiveness of Vaccines in a TB Dynamic*. International Journal of Engineering Research and Applications, v. 4, n. 6, pp. 233–238, 2014.
- Fronza, C. F., **Anselmo, J.**, Marques, J. L. B. *In Silico System for Early Diagnosis of Diabetes Mellitus Complications by Using Dynamic Pupillometry and Heart Rate Variability*. BIOMAT 2014 — International Symposium on Mathematical and Computational Biology, Poznan, Poland, 2014.
- **Anselmo, J.**, Suzuki, D. O. H., Marques, J. L. B. *Numerical Studies of the Influential Parameters of Electroporation Through Stretched Microcapillaries*. XXIV Brazilian Congress on Biomedical Engineering (CBEB), Uberlândia, Brazil, 2014.

Patents and Registrations

- **Anselmo, J.** *Electronic process using ultrasound technology for real-time volumetric measurement in liquid waste trucks*. Brazil, 2025. Invention patent, application no. BR 10 2025 001160 3, filed 22 January 2025 with INPI (Brazilian National Institute of Industrial Property). Funded by ENSA Tecnologia.
- **Anselmo, J.**; Anselmo, T. *Trademark registration for ENSA Tecnologia products and services*. Brazil, 2016. Registered trademark no. 925500895, INPI.

Peer Review and Teaching

Teaching Assistantship — Transducers Laboratory

2025, 2012

Federal University of Santa Catarina (UFSC) · Florianópolis, Brazil

Taught the *Transducers Laboratory* course for the Electrical and Electronic Engineering programs, in two separate assistantships.

Article Reviewer

2013–2014

American Medical Informatics Association (AMIA)

Reviewed submitted manuscripts against the association's review criteria; three reviews completed in 2014.

Undergraduate Thesis Committee Member — three committees

2014, 2017

UFSC and IFSC · Brazil

Theses examined: *Analysis of electrical, geometrical and chemical parameters in the application of nanosecond electric fields to biological cells* and *In vitro electroporation study: numerical simulation and experiments* (both UFSC, Electronic Engineering, 2017); *Epileptic seizure detection from electroencephalogram signals using the Wavelet transform* (IFSC, Telecommunications Systems, 2014).

Instructor — Biomedical Signal Processing

2013

V Minicourse on Biomedical Engineering in Practice · Brazil

Designed and taught a short course on biomedical signal processing.

Organizer

2012

IV Minicourse on Biomedical Engineering in Practice · Brazil

Professional Experience

Co-founder and R&D Director

2009–PRESENT

ENSA Tecnologia · Palhoça, Brazil

Lead the company's research and development function with a team of four, owning product concept and electronics design end to end — from microcontroller hardware to embedded algorithms — across healthcare, industrial automation, environmental monitoring and IoT. Recent work includes the ultrasound-based real-time volumetric measurement process for liquid waste trucks, filed as a patent in 2025 (see *Patents and Registrations*).

Selected products designed in this role:

- **ENSABiofeedback-100A** (2021) — microcontroller-based home-care device acquiring multiple physiological signals: photoplethysmography (PPG), electromyography (EMG) and galvanic skin response (GSR), for biofeedback assessment and training.
- **ENSAPlate** (2017) — optical character recognition and pattern classification using artificial neural networks, applied to vehicle license plate recognition for fuel pump automation.
- **ENSAIoT-200A** (2017) — IoT monitoring of volume, pressure, water flow and temperature, with real-time data transmission over a SaaS interface.
- **ENSAIoT-100A** (2016) — open-source microcontroller development board: analog and digital I/O, Bluetooth, Wi-Fi and serial (RS-232/RS-485) interfaces, LCD, touch keypad and electromechanical relay outputs.
- **ENSAMed-200S** (2019) and **ENSAMed-100B** (2011) — environmental sensing and volumetric measurement for fuel station pump sumps and tank interstitial spaces.

Earlier products: biometric access control with Ethernet and GSM/GPRS (2015), RFID-operated retail entertainment platform (2012) and a monitored residential alarm panel (2009).

Technical Skills

Programming and modeling: Python, Matlab, Octave, C/C++, Java; COMSOL Multiphysics, PSpice, Proteus

Embedded and electronics CAD: Microchip PIC, Atmel AVR, Espressif ESP; Altium Designer, KiCAD, Eagle

Instrumentation: Oscilloscope data acquisition, impedance spectroscopy, conductivity measurement

Scientific writing: LaTeX (IEEE templates), technical documentation

Open-source work: github.com/janioanselmo — **ISFAnalyzer** (oscilloscope waveform analysis),

ImpedanceAnalyze4294A (impedance spectroscopy GUI), **SearchArticle** (literature search), in Python

Languages

Portuguese: Native · **English:** Intermediate · **Spanish:** Basic